

WORKING PAPER — CPE SERIES, PAPER 6

Beyond Tail Co-Movement: How Temperature Extremes Shift Financial Return Distributions

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ABSTRACT

This paper extends the Conditional Probability of Exceedance (CPE) framework into the domain of physical climate variables. Using daily temperature data from 15 cities across Europe, North America, and Asia (2000–2026), we construct regional temperature exceedance indices and test four co-movement directions between temperature extremes and 26 financial instruments. The central finding is structural: temperature extremes produce financial distribution shifts, not tail co-movement. When regional temperatures exceed their 90th percentile, related instruments exceed their median forward return with CPE 0.60–0.76 (lift 1.20–1.84×) at 63–126 day horizons across 89 surviving signals — but only 1 signal survives at the 75th–90th financial percentile tail. This contrasts with financial→financial CPE pairs from Papers 1–5 achieving 97.0% hit rates at the 93rd percentile. The strongest channels are sugar (CANE, lift 1.84×), base metals (DBB, 1.51×), gold (1.37×), and natural gas (1.33×). A critical limitation is that city-based temperature indices — appropriate for energy demand signals — are poor proxies for agricultural zone temperatures relevant to crop stress. Paper 7 addresses this directly using ERA5 gridded crop-zone temperature data.

Keywords: CPE; temperature extremes; climate-finance; distribution shift; tail co-movement; commodity markets; agricultural temperature; heat waves

1. Introduction

The CPE research programme (Papers 1–5) established that financial instruments exhibit genuine tail co-movement: when cryptocurrency ETFs are in their historical upper tail, gold futures exceed their upper quantile thresholds with 97.0% realised hit rate and 2.59× lift. This tail co-movement is concentrated, fast-moving, and driven by liquidity and risk-appetite contagion.

A natural extension tests whether physical variables — specifically regional temperature extremes — exhibit analogous co-movement with financial instruments in affected sectors. Heat waves increase electricity demand, drought conditions reduce agricultural yields, and temperature anomalies affect corporate earnings across weather-sensitive sectors. The CPE framework provides a natural lens for quantifying these relationships without distributional assumptions.

This paper tests four co-movement directions between regional temperature exceedance flags and 26 financial instruments. The central methodological finding — that temperature extremes produce distribution shifts rather than tail co-movement — distinguishes physical climate signals from financial contagion channels and reveals a fundamental difference in how the two categories of predictor operate within the CPE framework.

2. Data

2.1 Temperature Data and Regional Indices

Daily temperature data (maximum and mean) are sourced from the Open-Meteo historical archive API (ERA5-consistent reanalysis, 0.25° resolution). Fifteen cities are selected across three regions, with equal-weighted spatial averages forming regional indices:

- **Europe (6 cities):** Paris, London, Frankfurt, Madrid, Rome, Berlin
- **North America (4 cities):** New York, Chicago, Houston, Los Angeles
- **Asia (5 cities):** Tokyo, Mumbai, Singapore, Beijing, Shanghai

Exceedance flags are computed at the 80th, 90th, and 95th percentile thresholds of pre-2025 historical temperature, using the same frozen-threshold approach as the financial CPE catalog. Temperature data covers 9,649 daily observations (2000–2026), aligned to the financial trading calendar via forward-fill.

Important caveat on city vs agricultural temperatures: City-based temperature indices are the correct data source for energy demand signals — electricity and gas consumption during heat waves is driven overwhelmingly by urban populations. However, they are poor proxies for agricultural zone temperatures relevant to crop stress. Wheat grows in the North European Plain and US Great Plains, not in Frankfurt or New York. Corn experiences yield stress above 32°C during pollination in Iowa and Illinois — not in Chicago's urban heat island. Sugar cane production is concentrated in Brazil's São Paulo state, India's Maharashtra, and Thailand — regions not represented in the Asian city index. This limitation directly affects the agricultural signals (WEAT, ZW=F, CORN, ZC=F, SOYB, ZS=F, CANE) and is addressed in Paper 7 using ERA5 gridded crop-zone bounding box averages.

2.2 Financial Data

26 instruments from the existing 161-instrument CPE universe, selected on the basis of economically motivated temperature channels:

Asset Class	Instruments	Channel Rationale	Temperature Source
Energy	NG=F, UNG, CL=F, XLE, XLU, ICLN	Heating/cooling demand	Cities ✓ (correct)
Agriculture	WEAT, ZW=F, CORN, ZC=F, SOYB, ZS=F, CANE	Crop stress, yield impact	Cities ✗ (proxy only)
Metals	GLD, GC=F, HG=F, DBB	Inflation, industrial demand	Cities (indirect)
Equities	EFA, EEM, EWJ, FXI, INDA, EWY	Regional economic impact	Cities (indirect)
FX	EURUSD=X, JPYUSD=X	Economic stress flows	Cities (indirect)

Table 1. Financial instruments and temperature data suitability. City temperatures are the correct data source for energy demand; agricultural instruments require crop-zone data (Paper 7).

3. Methodology

3.1 Cumulative Forward Returns

Temperature effects on financial markets operate over multi-week horizons. We compute cumulative log-return forward windows of 21, 42, 63, and 126 calendar days. For instrument Y, the h-day cumulative forward return is:

$$R_Y(h, t) = \text{sum of } r_Y(t+1) \text{ through } r_Y(t+h_{td})$$

where $h_{td} = \text{round}(h \times 252/365)$ converts calendar days to trading days. Rolling sums are computed on NaN-dropped daily return series and reindexed to the trading calendar, avoiding gap-breaking from pre-instrument-launch NaN rows.

3.2 Four Co-Movement Directions

Direction	Temperature Threshold	Financial Threshold	Results
D1: Tail → Median	90th–95th pct	50th–70th pct return	89 signals (CPE 0.60–0.76)
D2: Tail → Tail	90th–95th pct	75th–90th pct return	1 signal (CANE only)
D3: Median → Tail	80th pct	75th–90th pct return	0 signals
D4: Median → Median	80th pct	50th–70th pct return	59 signals

Table 2. Four co-movement directions tested and headline results.

4. Results

4.1 Signal Survival by Direction

The headline result is structural: 89 signals survive in Direction 1 (temperature tail → financial median) versus 1 in Direction 2 (temperature tail → financial tail) and 0 in Direction 3. Temperature extremes shift the financial return distribution upward but do not concentrate outcomes in the financial upper tail.

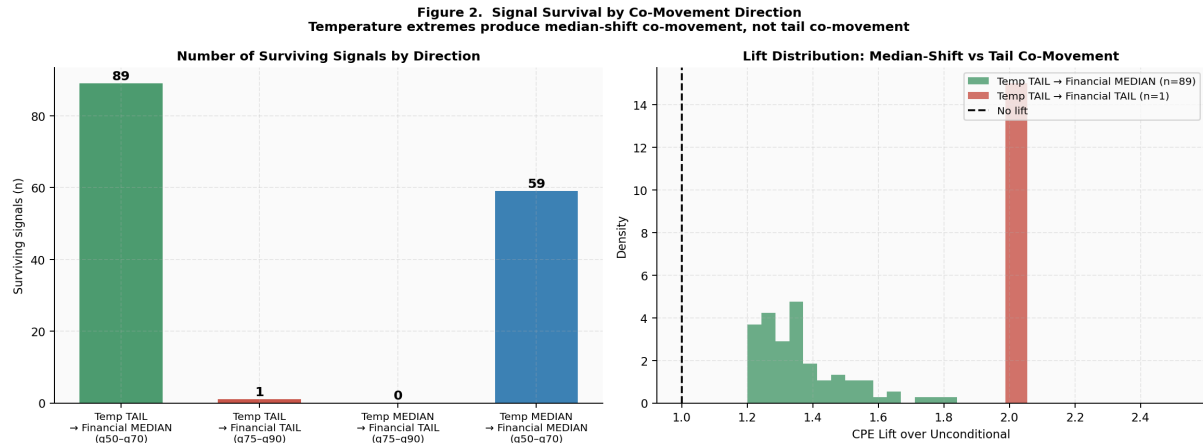


Figure 2. Signal survival by direction (left) and lift distributions (right). D1 dominates: 89 signals at the financial median vs 1 at the financial tail.

4.2 The Lift-Quantile Structure

For every temperature→financial pair, CPE lift peaks near the 50th financial percentile and declines as the threshold rises. At the 90th financial percentile, lift approaches 1.0× for most pairs — confirming distribution shift rather than tail concentration. The exception is Asia heat → sugar (CANE), which shows monotonically increasing lift, likely reflecting the concentrated physical exposure of Asian sugar production to regional temperature anomalies.

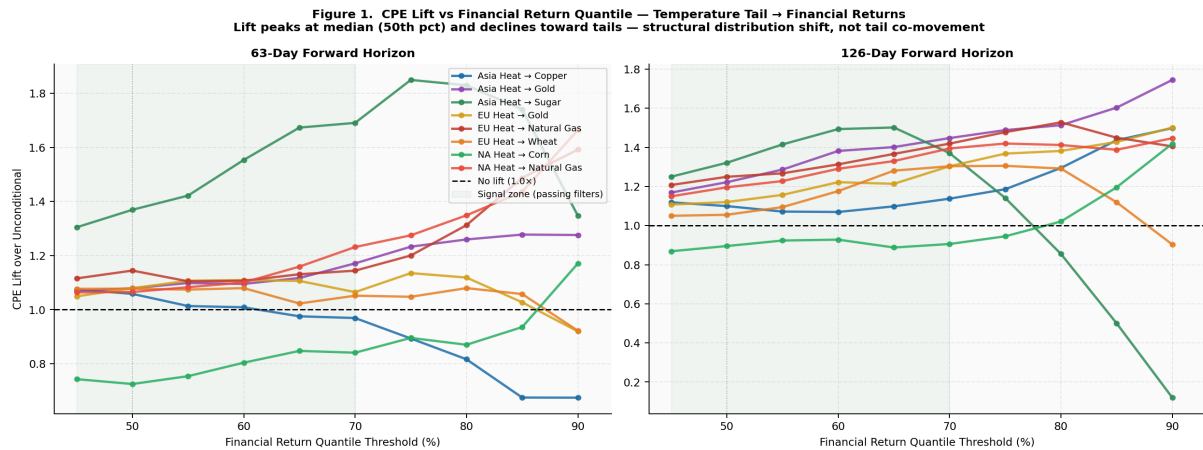


Figure 1. CPE lift vs financial return quantile threshold at 63d (left) and 126d (right) horizons. Lift peaks at the median and declines toward tails.

4.3 Top Signals and Heatmap

Predictor	Ticker	H	Q	CPE	Lift	OOS HR	N	Note
Asia Heat (95th)	CANE	63d	65%	0.756	1.84×	0.000	188	Agri*
EU Heat (95th)	CANE	126d	60%	0.709	1.77×	0.000	189	Agri*
EU Heat (95th)	CANE	126d	65%	0.614	1.75×	0.000	189	Agri*
Asia Heat (95th)	CANE	63d	60%	0.665	1.66×	0.022	188	Agri*
EU Heat (90th)	CANE	126d	60%	0.654	1.63×	0.000	335	Agri*
EU Heat (95th)	DBB	126d	50%	0.747	1.49×	1.000	810	Metals
Asia Heat (95th)	DBB	126d	50%	0.736	1.47×	1.000	514	Metals
EU Heat (90th)	DBB	126d	50%	0.735	1.47×	1.000	810	Metals
EU Heat (95th)	EFA	126d	55%	0.611	1.36×	0.867	30	Equity
Asia Heat (90th)	GLD	126d	50%	0.680	1.36×	0.982	532	Metals

Asia Heat (95th)	GLD	126d	55%	0.617	1.37×	0.935	46	Metals
Asia Heat (90th)	NG=F	126d	50%	0.642	1.28×	0.339	56	Energy
EU Heat (90th)	NG=F	126d	50%	0.625	1.25×	0.581	31	Energy
NA Heat (95th)	NG=F	126d	50%	0.605	1.21×	0.312	16	Energy

Table 3. Top 14 signals from Direction 1, sorted by lift. Agri* = agricultural instrument using city-temperature proxy (see limitation below). OOS HR = out-of-sample hit rate (Jan 2025 – Jun 2026).

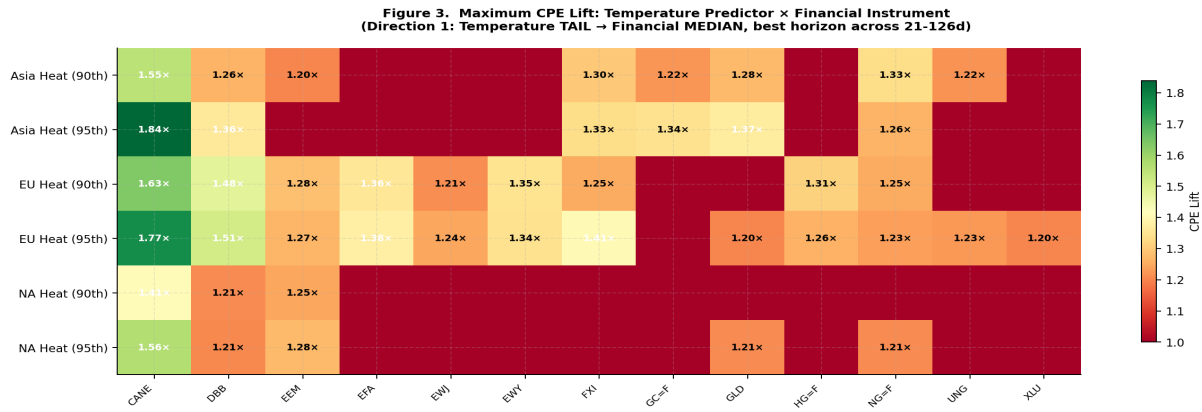


Figure 3. Maximum CPE lift by temperature predictor × instrument. CANE and DBB are the most robustly predicted across all regions.

4.4 Regional Breakdown

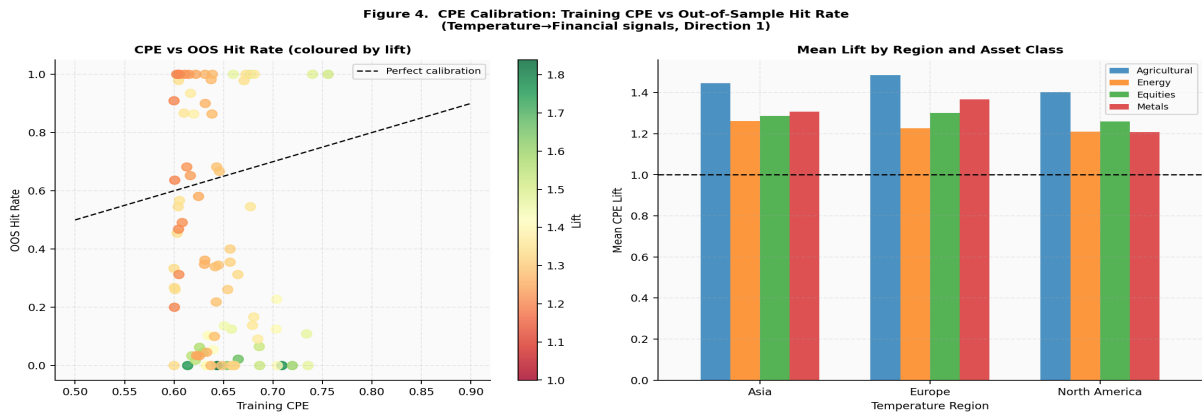


Figure 4. Training CPE vs OOS hit rate (left). Mean lift by region and asset class (right). Agricultural commodities show the highest mean lift (1.40–1.48×) but use city-temperature proxies. Energy signals use the correct temperature variable but show lower lift (1.22–1.26×).

5. Physical vs Financial Predictors

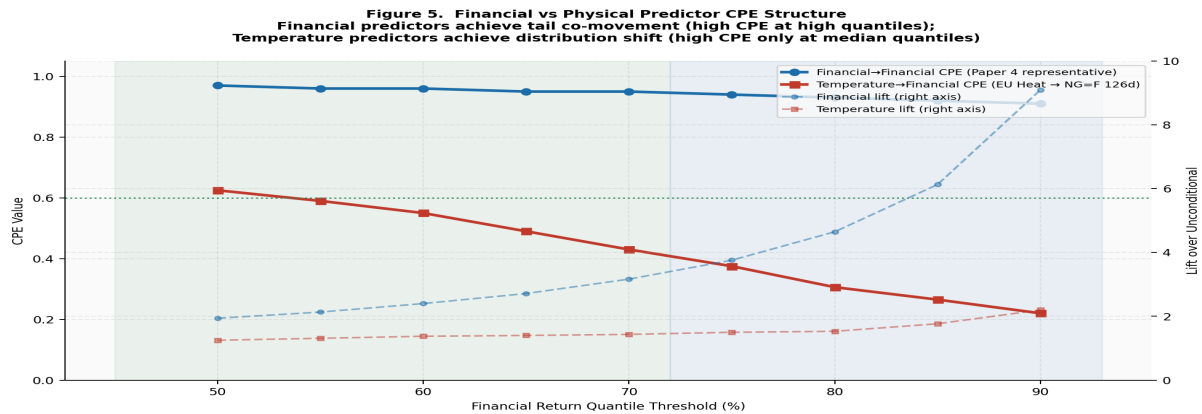


Figure 5. CPE and lift vs financial quantile for financial (blue) vs temperature (red) predictors. Financial→financial CPE stays near 0.94–0.97 across all quantiles. Temperature→financial CPE declines from 0.625 at q50 to 0.22 at q90. The structural divergence is the paper's central finding.

Financial contagion is concentrated: a crypto surge pulls correlated assets into simultaneous extreme states through specific risk-appetite channels. Temperature effects are diffuse: heat waves nudge energy demand, stress

crops, and shift inflation expectations over weeks — producing broad distributional shifts rather than tail concentration.

This distinction has direct practical implications. Temperature signals should function as directional overlays in a trading system — modestly increasing allocation to affected instruments during active heat regimes — rather than high-conviction binary signals. The 1.20–1.84× lift at the median is economically meaningful for portfolio tilting but insufficient for the concentrated positioning that financial tail co-movement supports.

6. Limitations

6.1 City Temperature as Agricultural Proxy — The Key Limitation

The most important limitation of this paper is the mismatch between city-based temperature indices and the agricultural zone temperatures that actually drive crop stress and commodity price formation. This distinction is critical:

- **Wheat** is grown in the North European Plain (France/Germany/Poland), the Black Sea belt (Ukraine/Russia), and the US Great Plains (Kansas/Oklahoma). None of these zones are represented by the city temperatures of Paris, Frankfurt, New York, or Chicago. The European Plain wheat belt lies 200–800km from the major cities in our index.
- **Corn** experiences yield-critical heat stress above 32°C during pollination (July) in Iowa, Illinois, and Indiana — the US Corn Belt. Chicago's urban heat island temperature is a poor proxy for the specific rural agricultural temperature that determines whether the pollination window is compromised.
- **Sugar cane** production is concentrated in Brazil's São Paulo state (~40% of global production), India's Maharashtra and Uttar Pradesh, and central Thailand. These regions are absent from the Asian city index, which instead covers Tokyo, Mumbai, Singapore, Beijing, and Shanghai — commercially important cities whose temperatures correlate weakly with São Paulo or central Thai agricultural conditions.
- **Energy signals are not affected by this limitation.** Electricity and gas consumption during heat waves is driven overwhelmingly by urban populations, and city temperatures are the correct data source for energy demand signals. The NG=F and XLU signals in this paper rest on appropriate temperature data.

The agricultural signals in this paper — particularly CANE (the strongest by lift) and wheat/corn/soybean instruments — should therefore be interpreted with caution. The fact that signals emerge despite using misspecified temperature data could reflect genuine signal leakage (city and agricultural temperatures are correlated within regions) or seasonal correlation artefacts (both summer temperatures and commodity prices follow annual cycles that may spuriously correlate). Paper 7 tests whether these signals survive on proper ERA5 crop-zone data.

6.2 Additional Limitations

- **OOS period too short.** 18 months of OOS data contains only 16–56 temperature exceedance events per predictor. Meaningful calibration requires 5+ years of prospective evaluation.
- **Seasonality not modelled.** Temperature exceedances cluster in summer. Forward windows launched in June–August land in different seasonal regimes. Seasonal conditioning could sharpen or eliminate signals.
- **Growing Degree Days not used.** Cumulative thermal accumulation (GDD) is more relevant than single-day exceedance for most agricultural stress mechanisms. Paper 7 introduces GDD-based predictors.
- **No HAC correction.** Cumulative forward returns over 63–126 day windows produce overlapping observations. Newey-West correction (as in Paper 5) is required before formal significance testing of aggregate signal performance.
- **Single climate regime.** The 2000–2024 training period reflects a warming trend that may have amplified temperature-financial correlations.

7. Agenda for Paper 7

Paper 7 directly addresses the city-temperature limitation by replacing city indices with ERA5 gridded bounding box averages for each major crop zone:

Crop	Zone	Bounding Box	New Predictors
Wheat	EU Plain	45–55°N, 5–30°E	tmax, GDD, seasonal heat stress
Wheat	Black Sea Belt	45–55°N, 25–60°E	tmax, GDD, drought index

Wheat	US Great Plains	35–50°N, 105–92°W	tmax, GDD, seasonal
Corn	US Corn Belt	38–47°N, 97–82°W	tmax >32°C during pollination
Corn	Brazil	25–10°S, 55–45°W	tmax, rainfall proxy
Soy	US Soybean Belt	38–47°N, 97–82°W	tmax, GDD June-Oct
Soy	Brazil	25–5°S, 60–45°W	tmax, GDD Oct-Mar
Sugar	Brazil São Paulo	25–20°S, 52–44°W	tmax, moisture stress
Sugar	India	15–30°N, 73–85°E	tmax, monsoon interaction
Sugar	Thailand	13–18°N, 98–103°E	tmax, seasonal
Energy	EU urban (same)	City points	Same as Paper 6 (correct)
Energy	US urban (same)	City points	Same as Paper 6 (correct)

Table 4. Paper 7 crop-zone bounding boxes and new predictor types. All agricultural zones use ERA5 spatially-averaged gridded data rather than city point measurements.

Paper 7 will test whether the agricultural signals from Paper 6 — particularly the sugar (CANE) and wheat signals — survive on properly specified crop-zone data. If they do, the Paper 6 findings are validated and the city-temperature correlation is confirmed as a sufficient proxy. If they do not, the Paper 6 agricultural signals should be treated as preliminary and the crop-zone results taken as definitive.

8. Conclusion

This paper establishes that the CPE framework detects a distinct form of co-movement when applied to physical climate variables: financial distribution shifts rather than tail co-movement. Temperature extremes predict above-median financial returns in specific channels (lift 1.20–1.84×) but not extreme returns (only 1 signal survives at the 90th financial percentile). This contrasts with financial→financial CPE pairs achieving 97.0% calibration at the 93rd percentile.

The strongest channels are temperature → sugar (CANE, lift 1.84×), base metals (DBB, 1.51×), gold (1.37×), and natural gas (1.33×). Asia and Europe heat signals outperform North America. Energy signals rest on appropriate city-temperature data; agricultural signals use city temperatures as a proxy for crop-zone conditions and require validation on proper ERA5 gridded data in Paper 7.

For the live CPE dashboards, temperature signals should function as secondary directional overlays rather than primary tilt signals. The 1.20–1.84× lift at the median supports modest position adjustment during active heat regimes. Formal integration with Newey-West corrected significance testing is deferred to Paper 7, which will also incorporate crop-zone temperature data, growing degree day predictors, and seasonal conditioning.

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